$C_{\phi d}^{(11)}$ i1_i2_ $\to$

$$\frac{1}{16\pi^2}\left(-\frac{2}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{d}}^{p}\right]\delta_{i1i2}+\frac{5}{108}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{d}}^{p}\right]\delta_{i1i2}-\frac{8}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{d}}^{p}\right]\delta_{i1i2}-\right.$$

$$\frac{2}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}+\frac{5}{36}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}-\frac{8}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}-$$

$$\frac{1}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}+\frac{5}{72}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}-\frac{4}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}-$$

$$\frac{1}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2}+\frac{5}{216}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2}-\frac{4}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2}-$$

$$\frac{8}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{u}}^{p}\right]\delta_{i1i2}+\frac{5}{27}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{u}}^{p}\right]\delta_{i1i2}-\frac{32}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{u}}^{p}\right]\delta_{i1i2}-$$

$$\frac{1}{27}g_1^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]\delta_{i1i2}-\frac{1}{18}g_1^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]\delta_{i1i2}+\frac{8}{135}g_1^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]\delta_{i1i2}-$$

$$\frac{1}{324}g_1^4\,\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}+$$

$$\frac{1}{162}g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}-$$

$$\frac{1}{324}g_1^4\,\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}+$$

$$\frac{1}{162}g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-$$

$$\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde{d}}^{i1}\right]\delta_{i1i2}-$$

$$\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-$$

$$\frac{1}{9}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{2,2,0}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]\delta_{i1i2}+\frac{1}{18}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]\delta_{i1i2}+$$

$$\frac{1}{162}g_1^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{d}}^{i1},m_1\right]\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{d}}^{i1},m_1\right]\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{d}}^{i1},m_3\right]\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{d}}^{i1},m_3\right]\delta_{i1i2}+$$

$$\frac{1}{162}g_1^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{d}}^{i2},m_1\right]\delta_{i1i2}-\frac{1}{324}g_1^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{d}}^{i2},m_1\right]\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{d}}^{i2},m_3\right]\delta_{i1i2}-\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{d}}^{i2},m_3\right]\delta_{i1i2}-$$

$$\frac{1}{9}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{2,2,0}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]\delta_{i1i2}+\frac{1}{18}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]\delta_{i1i2}+$$

$$\frac{1}{9}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{3,1,0}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}\right]\delta_{i1i2}+\frac{1}{9}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}\right]\delta_{i1i2}-$$

$$\frac{1}{4}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}\right]\delta_{i1i2}+\frac{1}{9}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}\right]\delta_{i1i2}+$$

$$\frac{1}{9}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]\delta_{i1i2}+\frac{1}{9}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]\delta_{i1i2}-$$

$$\frac{5}{12}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]\delta_{i1i2}+\frac{1}{3}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]\delta_{i1i2}+$$

$$\frac{1}{18}g_1^2\,\overline{a_u}^{pr}a_u^{pr}\,\mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]\delta_{i1i2}-\frac{1}{36}g_1^2\,\overline{a_u}^{pr}a_u^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]\delta_{i1i2}-$$

$$\frac{1}{18}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}^{p},\tilde{\mu}\right]+\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,2,-1}\left[m_{\tilde{q}}^{p},\tilde{\mu}\right]+$$

$$\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}^{p},\tilde{\mu}\right]-\frac{1}{18}g_1^2\,\overline{a_u}^{pr}a_u^{pr}\,\mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]\delta_{i1i2}-$$

$$\frac{1}{18}g_1^2\,\overline{a_u}^{pr}a_u^{pr}\,\mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]\delta_{i1i2}-\frac{1}{6}g_1^2\,\overline{a_u}^{pr}a_u^{pr}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]\delta_{i1i2}+$$

$$\frac{1}{3}g_1^2\,\overline{a_u}^{pr}a_u^{pr}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]\delta_{i1i2}+\frac{5}{36}g_1^4\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_1]\delta_{i1i2}-$$

$$\frac{1}{9}g_1^4\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_1]\delta_{i1i2}+\frac{5}{12}g_1^2g_2^2\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_2]\delta_{i1i2}-\frac{1}{3}g_1^2g_2^2\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_2]\delta_{i1i2}+$$

$$\frac{1}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,0}\left[\tilde{\mu},m_{\tilde{q}}^{p}\right]+\frac{7}{36}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{3,1,-1}\left[\tilde{\mu},m_{\tilde{q}}^{p}\right]-$$

$$\frac{1}{18}g_1^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{4,1,-2}\left[\tilde{\mu},m_{\tilde{q}}^{p}\right]+\frac{1}{72}g_1^4\,\mathrm{LF}_{2,1,1,-1}\left[m_1,m_{\tilde{d}}^{i1},\tilde{\mu}\right]\delta_{i1i2}+$$

$$\frac{1}{36}g_1^4m_1^2\,\mathrm{LF}_{2,1,1,0}\left[m_1,m_{\tilde{d}}^{i1},\tilde{\mu}\right]\delta_{i1i2}+\frac{1}{72}g_1^4\,\mathrm{LF}_{2,1,1,-1}\left[m_1,m_{\tilde{d}}^{i2},\tilde{\mu}\right]\delta_{i1i2}+$$

$$\frac{1}{36}g_1^4m_1^2\,\mathrm{LF}_{2,1,1,0}\left[m_1,m_{\tilde{d}}^{i2},\tilde{\mu}\right]\delta_{i1i2}+\frac{1}{72}g_1^4\,\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde{\mu},m_{\tilde{d}}^{i1}\right]\delta_{i1i2}-$$

$$\frac{1}{72}g_1^4m_1^2\,\mathrm{LF}_{2,2,1,-1}\left[m_1,\tilde{\mu},m_{\tilde{d}}^{i1}\right]\delta_{i1i2}+\frac{1}{72}g_1^4\,\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde{\mu},m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-$$

$$\frac{1}{72}g_1^4m_1^2\,\mathrm{LF}_{2,2,1,-1}\left[m_1,\tilde{\mu},m_{\tilde{d}}^{i2}\right]\delta_{i1i2}-\frac{1}{8}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,2,1,-1}\left[m_1,\tilde{\mu},m_{\tilde{q}}^{p}\right]-$$

$$\frac{3}{8}g_2^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,2,1,-1}\left[m_2,\tilde{\mu},m_{\tilde{q}}^{p}\right]+\frac{1}{4}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,0}\left[\tilde{\mu},m_1,m_{\tilde{q}}^{p}\right]+$$

$$\frac{3}{4}g_2^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,0}\left[\tilde{\mu},m_2,m_{\tilde{q}}^{p}\right]+\frac{1}{12}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{1,1,1,1,0}\left[m_1,m_{\tilde{d}}^{i1},m_{\tilde{q}}^{p},\tilde{\mu}\right]-$$

$$\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\left[m_1,m_{\tilde{d}}^{i1},m_{\tilde{q}}^{p},\tilde{\mu}\right]+$$

$$\frac{1}{12}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{1,1,1,1,0}\left[m_1,m_{\tilde{d}}^{i2},m_{\tilde{q}}^{p},\tilde{\mu}\right]-\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}$$

$$\mathrm{LF}_{2,1,1,1,-1}\left[m_1,m_{\tilde{d}}^{i2},m_{\tilde{q}}^{p},\tilde{\mu}\right]-\frac{1}{4}\tilde{\mu}^2\,\overline{y_d}^{ps}\,\overline{y_d}^{ri1}y_d^{pi2}y_d^{rs}\,\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{s},m_{\tilde{q}}^{p},m_{\tilde{q}}^{r},\tilde{\mu}\right]-$$

$$\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{i1},m_1,m_{\tilde{q}}^{p},\tilde{\mu}\right]-\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}$$

$$\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{i2},m_1,m_{\tilde{q}}^{p},\tilde{\mu}\right]+\frac{1}{18}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_1,m_{\tilde{d}}^{i1},m_{\tilde{d}}^{i2}\right]+$$

$$\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_1,m_{\tilde{d}}^{i1},\tilde{\mu}\right]+\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}$$

$$\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_1,m_{\tilde{d}}^{i2},\tilde{\mu}\right]+\frac{2}{3}g_3^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_3,m_{\tilde{d}}^{i1},m_{\tilde{d}}^{i2}\right]+$$

$$\frac{1}{4}\overline{y_d}^{si1}y_d^{pi2}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},\tilde{\mu}\right]+\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}$$

$$\left.\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu},m_1,m_{\tilde{d}}^{i1},m_{\tilde{q}}^{p}\right]+\frac{1}{24}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pi1}y_d^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu},m_1,m_{\tilde{d}}^{i2},m_{\tilde{q}}^{p}\right]\right)$$